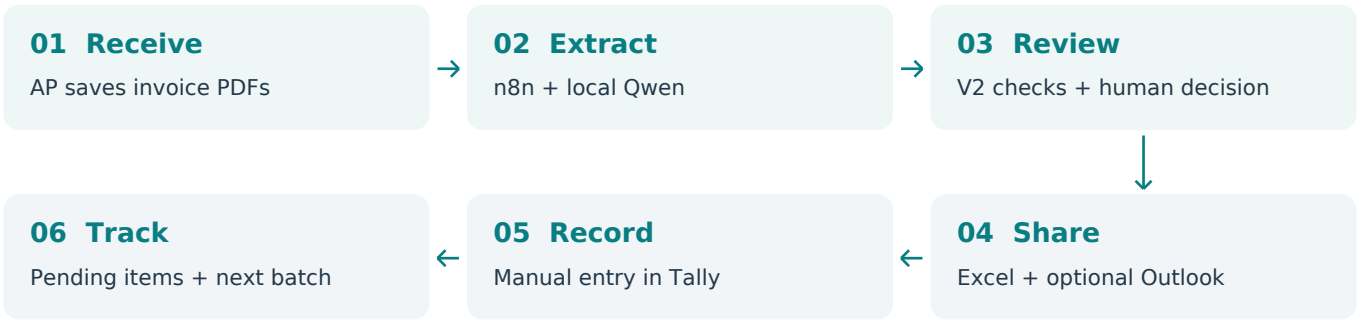
LOCAL AI · ACCOUNTS PAYABLE · TRAINING EDITION

Invoice Control V2

User manual & trainer guide

From an invoice received by email to a reviewed Excel pack, an Outlook handoff and a traceable accounting record.



What you will be able to do

Install the local components, run a batch, review invoices, approve or hold data, manage corrections, send a selected report through Classic Outlook and record the reference after manual Tally entry.

Includes: exact Windows folders and Docker configuration, a daily AP procedure, a training demonstration plan and 25 FAQs.

Edition: 9 October 2026 · Supplied V2 approval-and-tracking package, including Outlook Sender Fix 2.

Documented against the supplied application and workflow files. The latest Outlook sender correction still needs a successful check and send on the user's Windows PC.

READING MAP

Start here

Already running V1 or V2?

Keep the working installation. V1 remains available for your original training demonstration. Existing V2 users can go directly to the daily procedure on page 12; use setup pages only for a new PC or an incomplete installation.

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Who does what? The installer handles the initial PC setup. The AP operator receives and checks invoices; the reviewer authorises the current data version; the accountant makes the actual Tally entry. These are working responsibilities, not separate permissions enforced by this one-reviewer application.

PURPOSE & FEATURES

What this program solves

Invoices arrive in different vendor layouts. Manually retyping each one, spotting duplicates and remembering unresolved cases takes time. V2 converts readable invoice PDFs into a common register and keeps the review decision connected to the source document, batch and correction history.

Feature to show trainees	Practical result
Local AI extraction	Qwen reads extracted PDF text and proposes invoice fields, line items and a category.
Batch and invoice history	Each run logs file receipts; each invoice has a tracking ID and version history.
Checks after extraction	Required fields, arithmetic, possible duplicates and accounting balance are checked separately from the AI answer.
Evidence beside the data	Open the PDF, compare fields and review suggested debit and credit entries.
Human decision	Approve, hold, return for correction, close a rejection or identify a duplicate, with remarks.
Correction and carry-forward	Link a replacement to the original case. Open items remain visible in later batches.
Controlled Excel handoff	Create approved-entry, outstanding and batch workbooks; retain the exact report snapshot.
Optional Outlook + backup	Send one chosen report through Classic Outlook and back up the database with its documents.

Current scope

Readable-text PDFs, one invoice per PDF, ordinary INR supplier invoices, one company and one named reviewer. Tally posting, payments, OCR, mailbox downloading, automatic vendor reminders and PO/GRN matching are outside this release.

V1 can still demonstrate the simple extraction/report flow. V2 adds the review application and persistent history. Its report structure and email attachment behaviour differ from V1; see page 16.

TECH STACK & ADDRESSES

The technology, in simple terms

Component	What it does here
Windows + WSL 2	Windows is your desktop. WSL 2 supports Docker’s Linux containers.
Docker Desktop	Runs n8n and the V2 approval service in separate containers.
n8n	Runs the steps: receive files, read PDF text, call the model and save results.
Ollama + Qwen	Ollama runs qwen2.5:7b on Windows. Qwen extracts structured invoice facts.
Python + SQLite	The V2 application checks data and stores batches, versions, decisions and report history.
Excel + Classic Outlook	Excel is the manual-entry handoff. The Windows helper sends a selected report using Outlook.
Tally	The accountant enters and verifies the approved data manually.

Addresses you open in your Windows browser

Address	Purpose
http://localhost:5678	n8n: import and run the workflow.
http://localhost:8787	V2: review, decide, report and track.
http://localhost:11434	Ollama service check; this is not the approval screen.

Addresses used inside the n8n workflow

Ollama: `http://host.docker.internal:11434`
V2 service: `http://invoice-approval-v2:8787`

Why 5678 and 8787 are both correct

They identify different applications on the same PC. Inside a container, localhost refers to that container itself; it does not refer to your Windows desktop. The installer connects the two containers through Docker’s network. [12]

ONE-TIME SETUP · NEW PC

Install the prerequisites

Skip this page on the existing working PC. On a new training PC, complete the following in order. Use the official links on page 27 for current installer requirements.

01 Prepare Windows and WSL 2

Use supported 64-bit Windows with hardware virtualisation enabled. If WSL is absent, open PowerShell as administrator, run `wsl --install`, then restart. Check with `wsl --version` and `wsl -l -v`. You do not need to work in Ubuntu each day. [1]

02 Install and start Docker Desktop

Select the WSL 2 backend and Linux containers. Start Docker Desktop and wait for the engine to run. The n8n image is downloaded from Docker Hub through the compose command on page 7; do not also create a second n8n container manually. [2]

03 Install Ollama for Windows

Start Ollama from the Windows Start menu. Open a new PowerShell window and download the supplied model using the command below. The model tag must match the workflow settings. [3, 4]

04 Have the working files and desktop tools

Keep the supplied V2 ZIP and the 20-invoice pack. Use a browser, a PDF reader and an Excel-compatible spreadsheet viewer. Classic Outlook with your sending account configured is needed only for the optional email feature.

```
ollama pull qwen2.5:7b
ollama list
ollama run qwen2.5:7b "Reply with OK only"
```

Hardware: the trainer's stated PC has 24 GB RAM and an RTX 5050 with 8 GB VRAM. Use 16 GB or more RAM as a practical starting recommendation for a similar workshop, not a performance guarantee. The 7B model download is about 4.7 GB; runtime memory, Docker and files require extra capacity. A smaller model needs its own accuracy test. [4]

Is the database free?

Yes. SQLite is included and has no separate database licence fee or server installation. The overall setup still uses your PC and any applicable Outlook, Docker or n8n licences. Local Qwen calls do not incur a cloud-model API charge. [6–8]

ONE-TIME SETUP · WINDOWS

Create the files and folders

Use File Explorer. These are the paths for the supplied installation; a new trainee may substitute their own Windows base folder consistently. Keep the existing V1 folders and workflow.

Windows location	Purpose
C:\Jagjeet Singh\n8n\compose.yaml	Existing n8n Docker configuration. Run its compose commands from this folder.
C:\Jagjeet Singh\n8n-files	Windows folder mounted into n8n as /files.
...\n8n-files\invoice_demo	Original V1 training files; continue to use them for the V1 demonstration.
...\n8n-files\invoice_demo_v2	Extract the V2 package here. SETUP_V2.cmd must be directly inside this folder.
...\invoice_demo_v2\input	Copy only the PDF invoices for the V2 run here. Subfolders are not scanned.
...\invoice_demo_v2\archive	V2's retained source and replacement PDFs. Do not rename or remove these files.
...\invoice_demo_v2\output	Dated Excel snapshots. The outlook subfolder holds helper requests and receipts.
...\invoice_demo_v2\backups	Full V2 backup ZIP files. Copy important backups to a second location.

Other files in the V2 folder

Use: SETUP_V2.cmd once; START_V2.cmd, STATUS_V2.cmd and BACKUP_V2.cmd as needed. **Import:** Ready_To_Import_V2.json after setup. **Sign in:** read LOCAL_LOGIN_PRIVATE.txt privately. **Read:** START_HERE and the Outlook fix notes.

Leave in place: app, scripts, compose.v2.yaml and .env. The sample_invoices folder holds the training pack and answer keys; reference_v1 preserves the supplied V1 references. The outlook_bridge folder contains the Windows helper.

Two easy folder mistakes to avoid

Do not extract to invoice_demo_v2\invoice_demo_v2. Show file-name extensions in Explorer so compose.yaml is not accidentally saved as compose.yaml.txt. The input folder is not emptied automatically; archived copies are separate from the originals you copied in.

ONE-TIME SETUP · COMPOSE

Connect Docker to the Windows folder

The following reproduces the supplied n8n configuration. **Existing working users: keep your current compose.yaml.** For a new PC, save this as **C:\Jagjeet Singh\n8n\compose.yaml** after creating that folder and the n8n-files folder.

```
services:
  n8n:
    image: n8nio/n8n
    container_name: n8n
    ports:
      - "5678:5678"
    restart: unless-stopped
    environment:
      - N8N_RESTRICT_FILE_ACCESS_TO=/files;/home/node/.n8n-files
    volumes:
      - n8n_data:/home/node/.n8n
      - "C:/Jagjeet Singh/n8n-files:/files"
volumes:
  n8n_data:
    external: true
```

The Windows folder on the left of the mount is visible as **/files** inside n8n. Therefore the V2 input path is **/files/invoice_demo_v2/input** in the workflow. Quotes preserve spaces in the Windows path.

New PC only: run these in PowerShell, one line at a time

```
cd "C:\Jagjeet Singh\n8n"
docker volume create n8n_data
docker compose pull
docker compose up -d
docker ps
```

The volume command is needed because this compose file declares n8n_data as external. Open **http://localhost:5678** and finish n8n's first-use owner login. Docker Compose creates the network that the V2 installer later detects. [5, 9]

Fresh local-only installation: you may use `"127.0.0.1:5678:5678"` for the port line to limit browser access to this PC. The supplied original uses `"5678:5678"`. Record the working n8n version after rehearsal and avoid image updates immediately before training. [10]

ONE-TIME SETUP · APPROVAL APP

Set up V2 and its database

01 Check the existing services

Docker Desktop, the container named n8n and Ollama must be running. The Windows n8n-files folder must already be mounted to /files as shown on page 7.

02 Run SETUP_V2.cmd

Double-click it inside invoice_demo_v2. It checks the mount, detects the existing Docker network, creates the V2 working folders and starts the approval container. The first run may download the Python image.

03 Keep the generated connection files

Setup creates .env with local configuration, LOCAL_LOGIN_PRIVATE.txt with reviewer access, and Ready_To_Import_V2.json with the configured V2 connection. Use the reviewer name entered during setup; the script's suggested name is jagjeet.

04 Open the review application

Go to http://localhost:8787 and sign in using the generated reviewer credentials. This login is separate from your n8n owner login. Use START_V2.cmd for later starts; do not repeat installation as a daily step.

Where is the history kept?

Store	What it contains
invoice_v2_data Docker volume	The V2 SQLite database, created automatically. Internal file: /data/invoice_control_v2.db.
Windows invoice_demo_v2 folder	Source archive, reports, email-job files, backups and application files.
n8n_data Docker volume	n8n's own saved workflows and settings; separate from the V2 database.

No separate Windows Python, SQLite download, database server, SQL account or SQL knowledge is required. SQLite keeps batch receipts, invoice versions, decisions, exports and email records. Normal stop/start preserves this data.

Choose the company before the first batch

The demo defaults to the fictional Northstar Precision Components Pvt Ltd. Set the intended company in Demo Settings before loading any invoices. This workspace does not allow changing company once records exist. Keep training and real processing in separately configured workspaces; ask the maintainer to set this up.

N8N WORKFLOW

Import the JSON and run a first batch

01 Create a separate workflow

Open http://localhost:5678. Create a new workflow and use the workflow menu’s Import from File option. Select invoice_demo_v2\Ready_To_Import_V2.json, generated by setup. Name and save it clearly as V2. Keep V1 as a separate saved workflow. [5]

02 Check Demo Settings

Confirm the exact model tag qwen2.5:7b, Ollama base URL http://host.docker.internal:11434 and approval URL http://invoice-approval-v2:8787. Confirm the company. A blank as-of date uses the current date; use a training date deliberately when appropriate.

03 Prepare a small first lot

Copy a few known, selectable-text PDFs into invoice_demo_v2\input. One invoice may span several pages, but each PDF must represent one invoice. Keep each file under 25 MB. Copy only PDFs, not the answer-key CSV or JSON.

04 Execute once and wait

Use Execute workflow for the manual run. n8n registers and archives each PDF, extracts its text, calls local Qwen, saves the result and finishes the batch. Model speed varies; an active call is not automatically a failure.

05 Check the result in V2

Open http://localhost:8787, select Refresh, then Batch history. Check Received, New, Duplicate receipts, Completed and Errors. Open Invoice register and Review on an invoice. A completed extraction still requires approval.

What should you see after a normal run?

An invoice register with tracking IDs; the batch record and file receipts; source PDFs retained in archive; a Batch Summary and an All Outstanding workbook. The Approved Entries workbook is created later, after human approvals.

Use the generated file: workflows\Local_Invoice_V2_Workflow.json is the reusable template. Its placeholder connection is not the file to import for normal operation. Ready_To_Import_V2.json contains a local API token; keep it private and redact it before sharing.

First-run acceptance: compare one PDF to its extracted figures, verify one accounting entry balances, show a held item in Outstanding, and create an approved report from a reviewed item. Test Outlook separately using a recipient you intend to contact.

TRAINER REFERENCE

Get workflow JSON from an AI assistant

ChatGPT or Gemini can help create or revise an n8n workflow JSON. Give the assistant the actual environment and an example of the intended result. A valid JSON file is only the starting point: it must also match the installed n8n nodes and this application's API.

Provide these facts

Provide the n8n version, Windows/Docker paths, model name/URL, invoice fields, error handling and report requirements. For V2, ask the maintainer for the API contract (endpoints and expected fields), or attach app/server.py with the redacted workflow. Use fictional or authorised, redacted examples. Remove API tokens and passwords.

A useful prompt to copy

"I run n8n locally in Docker on Windows. My Windows invoice folder is mounted as /files. Ollama runs on Windows at http://host.docker.internal:11434 with qwen2.5:7b. I attach my redacted working workflow and API contract or server code. Create an importable n8n JSON for the requested change, compatible with n8n version [insert version]. Preserve the existing V2 batch IDs, error paths, version history and duplicate controls. Use one readable-text invoice PDF per file. Keep secrets as placeholders and identify all settings I must supply. Explain what changed, required files and a five-invoice test. Do not invent API endpoints, credentials, OCR or Tally posting. My requested change is: [describe it]."

From an AI answer to a usable file

01 Save the actual JSON

Download the .json attachment. If the answer is pasted text, copy only the JSON into a plain-text editor and save as UTF-8 with a .json extension, without Markdown fences or explanatory prose.

02 Import as a new workflow

Keep the known working version. Inspect nodes, URLs, paths, credentials, recipient settings and the error branch before executing.

03 Test known invoices

Compare output against the PDF and trainer answer key. Test an extraction failure, an exact duplicate and a correction as well as a clean invoice. Only then use a larger lot.

For this V2 use case: use the supplied setup-generated JSON first. A freshly invented workflow cannot create a compatible approval database, UI and Outlook bridge by itself. Those are separate files in the V2 package.

KEEP THIS PAGE NEAR THE PC

Simple command reference

Open **Windows PowerShell** for the commands below. Run one line at a time. These checks do not require you to learn Linux. Use command files by double-clicking them in File Explorer.

Command	What it tells you / does
`wsl --version`	Shows installed WSL version.
`wsl -l -v`	Lists WSL distributions and whether they use WSL 1 or 2.
`docker --version`	Confirms the Docker command is installed; it does not prove the engine is running.
`docker ps`	Lists running containers. Expect n8n and the V2 approval service after setup.
`docker ps -a`	Also shows stopped containers.
`docker start n8n`	Starts an existing stopped n8n container.
`docker logs --tail 30 n8n`	Shows recent n8n messages for troubleshooting.
`docker stats --no-stream`	Shows one snapshot of container CPU and memory use.
`docker exec n8n ls /files`	Checks the mounted folders visible inside n8n.
`ollama list`	Shows downloaded models. Check the exact qwen2.5:7b tag.
`ollama ps`	Shows loaded models. An empty list while idle is normal.
`ollama run qwen2.5:7b`	Opens a local model chat for a quick test. Type /bye to leave.

The V2 shortcuts

In invoice_demo_v2: START_V2.cmd starts; STATUS_V2.cmd checks; STOP_V2.cmd stops; BACKUP_V2.cmd backs up. Stop only after current processing has finished.

In outlook_bridge: CHECK_OUTLOOK_V2.cmd tests account selection; START_OUTLOOK_V2.cmd processes mail requests while its window stays open. Close the running helper before the check.

Normal operation is simpler than setup

Start Docker Desktop and Ollama from Windows. Do not run a second ollama serve while the Windows Ollama app is already serving. Do not delete Docker volumes or use cleanup/prune commands to cure an ordinary connection error.

OPERATOR PROCEDURE · PART 1

Daily AP work: receive and process

The operator starts in the mailbox and finishes with every invoice in a known state. Receiving email and downloading attachments are manual steps in this release.

01 Start the workspace

Start Docker Desktop and Ollama. Open n8n on 5678 and V2 on 8787. If necessary, start the existing n8n container and double-click START_V2.cmd. Keep Classic Outlook/helper for the later email step.

02 Download and inspect the invoice

Save the attachment from the vendor email. Confirm it opens, belongs to the intended company and contains one invoice. Try selecting its text. A scan needs separate OCR; a PDF with several invoices must be split before input.

03 Copy the PDFs into V2 input

Use C:\Jagjeet Singh\n8n-files\invoice_demo_v2\input. Keep a recognisable filename. Do not place the same lot in the V1 folder when demonstrating V2.

04 Run the V2 workflow once

Open the saved V2 workflow in n8n and execute it. Avoid overlapping runs. The application creates the batch record and copies source files into archive before extraction.

05 Reconcile the batch

In V2, Refresh and open Batch history. Investigate every error and duplicate receipt. Twenty received files do not necessarily mean twenty new liabilities: an unchanged repeat is recorded as a receipt and skipped.

06 Open each record for review

Go to Invoice register. Search by supplier, invoice number, file or tracking ID. Use the status and original-batch filters, then select Review. Follow pages 13–14 for the decision.

Keep input manageable after checking the run

Input files remain where you copied them. After confirming the files are archived and their batch is accounted for, move completed input copies to your own processed folder outside input. Keep any PDF needed for an extraction retry. Do not reorganise or delete V2’s archive.

If interrupted: confirm the n8n execution has stopped, inspect the batch, then use Close interrupted run with a reason before retrying after the fault is corrected. A retry does not mean that the first attempt created no records.

OPERATOR PROCEDURE · PART 2

Review the source and the entry

A successful model response is a draft extraction. The reviewer confirms both the printed invoice facts and the accounting treatment.

Screen/tab	What to check
Open PDF	Use the source beside the form. Check all pages of a long invoice and select the replacement document when relevant.
Invoice details	Supplier, invoice number, dates, buyer, currency, GSTIN, taxable value, each tax, other charges, round off, total and line items.
Accounting entry	Correct Tally company, voucher type/date, exact ledger names, cost centre, narration and balanced debit/credit amounts.
Decision	Read every finding. Resolve blocking items; explicitly acknowledge review exceptions with a supported reason.
History	Read Document & data versions and Decision trail. Earlier source documents and saved data remain available.

If the extraction is wrong but the PDF is right

Edit the verified fields. If you change amounts or category, also inspect and update **Accounting entry**; the journal is not automatically rebuilt by a field edit. Enter a **Reason for correction** and select **Save correction**. Review the new version and its checks before approving.

Understand the two kinds of findings

Blocking: missing essential facts or amounts, a printed-total mismatch, unresolved duplicate, failed extraction or an invalid/unbalanced accounting entry. Approval waits until the underlying problem is resolved.

Review: missing due date/GSTIN, unusual dates, partial line detail, quantity/rate differences, capex, deposits, professional services or high value. Each requires a human judgment and remarks. Default high-value review applies above INR 100,000; this backend setting is not a legal threshold or a screen control.

Do not make the source say something it does not

A blank tax field is not automatically zero. If the PDF clearly establishes that tax is not applicable, document that review before entering zero. If the vendor’s printed total is wrong, return it for correction; do not alter evidence merely to make checks pass.

GSTIN checks cover format, not a live government validation. Ledger suggestions do not come from your Tally master. Credit notes, foreign currency, TDS/split payables and inventory posting need a separate process.

Long PDFs: the workflow sends only the first 24,000 extracted text characters to the model and flags truncated text. Check completeness of all pages and line items before approval.

OPERATOR PROCEDURE · PART 3

Choose a decision and record the action

Action in Decision	When to use it	What happens next
Approve for manual entry	Source and accounting entry are checked; all blocks cleared.	Tick review exceptions, confirm company/ledgers/tax treatment and enter remarks. Approve version makes this version eligible for export.
Put on hold	A temporary dependency is unresolved, such as receipt confirmation.	Remains outstanding. Record what is needed, who will act and when to follow up.
Return for correction	The supplier must send a corrected invoice or supporting information.	Remains outstanding. Contact the supplier manually and later link the replacement.
Reject & close	The invoice is permanently invalid or cancelled.	Leaves outstanding, but remains in the register and history. Reopen for review if circumstances change.
Close as duplicate	You have identified the retained original invoice.	Choose that original and record why the copy is closed. Check the retained record separately.
Mark manually recorded	The approved entry is actually entered and verified in Tally.	Enter the real voucher reference. The record becomes Manually recorded and ordinary edits are locked.

Use remarks that another person can act on

Example of a useful hold remark

“Awaiting goods-receipt confirmation for invoice AB-104. Owner: stores supervisor. AP to call on 12 Oct. Release for review after signed receipt is received.”

Owner and target date are text in the remarks; the application does not assign tasks, schedule calls or send reminders.

How to read the register status

Processing means work is underway. **Processing problem** means it needs repair/retry. **Awaiting approval** means a review is still due. **On hold** and **Returned for correction** stay open. **Approved · awaiting entry** is ready for accounting handoff. **Manually recorded** is the user’s posting confirmation. **Rejected · closed** is a retained closed case.

Age is days since the invoice first reached V2. **Outstanding** means unfinished processing or recording, not an unpaid supplier balance. Approval is not payment authorisation.

OPERATOR PROCEDURE · PART 4

Corrections and the next lot

A corrected vendor PDF arrives

01 Process the new document

Put the replacement PDF into input and run a new batch. It appears as a new, unapproved record and may be flagged as a possible duplicate.

02 Link it to the original case

Open the old outstanding invoice. Choose Use corrected invoice, select the newly processed unapproved record, enter the reason and record the decision. If the old case is Rejected · closed, first use Reopen for review.

03 Review the new version

The original tracking ID receives the next data/document version; the temporary replacement record closes. Both documents remain in history. Recheck fields and accounting entry, then approve the current version.

04 Use a current report

The former approval does not carry over. A report containing a changed version is marked superseded; create a current approved export after fresh approval and stop using the earlier file.

Situation	How V2 treats it
Same unchanged PDF again	Recognises the content fingerprint and logs a duplicate receipt. Changing the filename does not make it new.
Changed readable PDF or reissued invoice	Checks supplier/invoice/company/financial-year identity. A candidate duplicate blocks approval until reviewed.
Earlier extraction failed	The same PDF can be retried after fixing the fault, as a new version of the same invoice record.
Already recorded in Tally	Use Record correction follow-up to retain the note. Resolve the real accounting correction separately; this does not amend Tally or reopen an ordinary posting queue.

Carry-forward happens in the database

Unresolved Hold, Return and Awaiting approval cases stay visible without copying their PDFs again. The next batch’s Opening Outstanding sheet captures the position at its start. A later duplicate can also reopen an approved, unrecorded case and supersede its export.

REPORTS & MANUAL TALLY ENTRY

Your final Excel handoff

Use **Reports & email → Create Excel report**. Each file is a dated snapshot with a unique V2 name, so it does not overwrite the original V1 reports.

Report	Contents and use
Approved entries for manual Tally entry	Approved Invoices, Accounting Entries, Read Me, Status Totals and Invoice Line Items. Create explicitly after review; use for the accounting handoff.
All outstanding invoices	Outstanding, Read Me, Status Totals and Invoice Line Items. Follow up open cases across batches. Excludes Manually recorded and Rejected · closed.
Batch summary	Batch Summary, File Receipts, Opening Outstanding, Current Invoice Register, Read Me, Status Totals and Invoice Line Items. Reconcile the lot and review its history.

Batch completion creates Batch Summary and Outstanding automatically. Generate a fresh report after decisions when you need current status. Opening Outstanding stays fixed to the batch start; Current Invoice Register reflects the time the report is generated.

From approved report to a real accounting record

01 Create and check the report

The approved export reserves its included invoice versions. Download and, if required, email that report before marking the invoices recorded.

02 Enter and verify in Tally

Use the intended Tally company and reviewed ledger names. The workbook is a reference for manual entry; V2 does not import it into Tally or verify the resulting voucher.

03 Confirm the entry in V2

Open the invoice → Decision → Mark manually recorded. Enter actual company, voucher type, voucher number and voucher date, confirm verification and select Confirm recorded.

Why a second approved export can omit the invoice

Its current version is already reserved in the first export. Re-download that snapshot. Release its reservation with a reason only if the entry has not been recorded and the earlier file will no longer be used. Editing a record requires fresh approval and makes earlier exports unsuitable for posting.

If you stop at approval and tracking: these Excel reports and retained PDFs are your final deliverables. Tally work can remain completely manual. Source links in reports point to the signed-in local app; they do not become public links for an email recipient.

OPTIONAL WINDOWS EMAIL HELPER

Send a report through Classic Outlook

01 Check the current helper once

Use the latest Sender Fix 2 files in outlook_bridge; these include the earlier startup-path fix. Close any running helper, open Classic Outlook, then run CHECK_OUTLOOK_V2.cmd. Look for PASS for the exact intended sender account. This check creates and discards blank messages; it does not send or process the queue.

02 Start the sender helper

Close the check window. Run START_OUTLOOK_V2.cmd and leave its window and Classic Outlook open. Setup does not start this helper. Starting it can process a fresh queued request, so check that any queued request is still intended.

03 Request one report email

In Reports & email, choose the current exported report and Email via Outlook. Enter the exact configured sender address, To addresses separated by semicolons and optional Cc. Confirm the recipients and send once.

04 Verify the result

Refresh the email queue, read the helper message and check Outlook Sent Items or Outbox. Send approved reports before Mark manually recorded: the application rechecks that every included invoice is still approved and eligible.

Queue status	Meaning / next action
queued	Waiting for a helper result. Check the helper is running and the job is still fresh.
submitted	Outlook accepted the send request. This is not proof of recipient delivery.
error	Read the error detail. A sender-selection failure saying “No send was attempted” occurred before sending.
unknown	Outcome is uncertain. Check Sent Items, Outbox and Drafts before requesting another email.

What is actually attached?

One selected Excel workbook and a generic explanatory email body. V2 does not automatically attach three separate workbooks, all source PDFs or a customised vendor message. Each request expires after three minutes and is not automatically retried.

Current desktop test status: Sender Fix 2 is included, but a successful sender check and real delivery still need confirmation on this PC. CHECK validates account selection, not mail-server access. The supplied bridge supports Classic Outlook, not the new Outlook application.

OPERATOR PROCEDURE · PART 5

Finish the day with clear communications

The objective is that approved work reaches the accountant and every unresolved invoice has a clear next action. A decision in the screen does not itself send a message.

Who needs what?	What the AP person does
Accountant: approved invoices	Create the current Approved Entries report, check it and use Email via Outlook to send that one workbook to the intended internal recipient. Then the accountant enters it and the actual voucher reference is recorded.
AP / reviewer / procurement: open actions	Generate a fresh All Outstanding report. Review reasons, owners and target dates written in remarks. Email the report internally if needed and make the follow-up calls manually.
Vendor: correction requested	Use Returned for correction in V2 with the exact issue. Compose a separate targeted Outlook message to the vendor, describing the correction and attaching only the relevant document or excerpt.
Vendor / manager: rejection explained	Use Reject & close with a clear reason. Review the record in Invoice register or filter the Current Invoice Register sheet in a batch report. Compose the appropriate targeted message manually.
Next shift or next lot	Leave unresolved records open, note the action taken and next follow-up, and create a full backup. Corrected invoices can be processed in a later batch and linked to the original.

Example: invoice total does not add up

The reviewer checks the PDF and confirms that its printed grand total is wrong. AP selects **Return for correction**, records the discrepancy and expected action, and emails the vendor manually. The invoice remains outstanding. When the corrected PDF arrives, AP processes it in the next lot, links it using **Use corrected invoice**, obtains fresh approval and includes it in a new approved export.

What “disapproved” means in practice

Use Hold for a temporary dependency, Return for a correction still expected, or Reject & close for a finished invalid case. Closed rejections are intentionally absent from Outstanding, but remain in the register, batch reporting and history.

There is no dedicated rejected-only workbook, automatic supplier routing, call scheduler or editable custom email body in V2. Use captured reasons to guide the manual follow-up. Keep multi-vendor internal reports within the intended internal group.

DAILY CARE & RECOVERY

Back up and preserve the record

01 Create a full V2 backup

With V2 running, choose Backup & restore → Create full backup, or double-click BACKUP_V2.cmd. Wait for completion and find the ZIP in backups.

02 Keep a second copy

Copy the ZIP to a separate protected disk or approved storage location. A copy on the same failing PC is not a sufficient recovery plan.

03 Preserve the additional setup items

Keep the supplied package, .env, LOCAL_LOGIN_PRIVATE.txt, n8n workflow exports and any unprocessed input separately and privately. The full V2 backup does not include these items or the n8n_data volume.

04 Close the day cleanly

Check active runs and mail requests first. Close the Outlook helper when no more sends are intended. STOP_V2.cmd stops the approval application without deleting its records. Normal restart does not erase the database.

Included in the V2 backup	Outside that backup
SQLite database snapshot	n8n workflows/settings and n8n_data volume
Linked archive PDFs	Unprocessed PDFs remaining only in input
Output reports and helper files	Application code, .env and private login file
Manifest used to check the backup	A separate backup of Windows, Outlook or Tally

Restore only when recovery is actually needed

Stop the V2 n8n execution and close the Outlook helper. Place the chosen ZIP in backups and run **RESTORE_V2.cmd**. Follow its filename prompt and deliberate **RESTORE V2** confirmation. The script stops V2, validates the backup, preserves a pre-restore backup and replaces the V2 database/archive/output with the selected snapshot. It restarts after success; if it fails, read the message before continuing.

A restore replaces the current V2 state

It is not a merge with newer work. Old mail jobs are quarantined instead of replayed. Reconcile Outlook before creating fresh send requests. Copying only the Windows invoice_demo_v2 folder is not a complete backup because the live database is in its Docker volume.

TRAINER RUN OF SHOW

Run a useful training demonstration

Use a training company and fictional invoices. Rehearse with the installed software versions before the class. Keep a separate V1 workflow available to show the original simple process first.

Demonstration	What the trainee should notice
1 · Compare V1 and V2	V1 shows extraction and reports; V2 adds retained batches, decisions, corrections and manual-entry tracking.
2 · Show the 20 layouts	The pack covers different fictional vendors and designs. PDFs contain selectable text; it is not an OCR benchmark.
3 · Process and reconcile	Run the lot and show file receipts, invoice IDs, source PDFs, automatic batch report and outstanding report.
4 · Review contrasting cases	Compare a clean invoice with PDF 15 (printed total INR 700 too high), PDF 18 (INR 15,000 refundable deposit), PDF 19 (no due date), and PDFs 07/20 (equipment/furniture review).
5 · Make the decisions visible	Approve a verified case; hold another; return PDF 15. Explain the difference between a blocking error and a review exception.
6 · Demonstrate history	Re-copy a processed PDF to show an exact duplicate. Use a clearly labelled corrected training PDF to demonstrate replacement and fresh approval.
7 · Show the final handoff	Create/download approved and outstanding reports. Demonstrate optional email to an intended test recipient, then show manual Tally reference capture using a real training voucher.
8 · Start another lot	Show opening outstanding, unresolved cases and the audit trail. Finish with a backup.

The initial pack contains no deliberate duplicate. Use expected_results.csv and expected_invoice_data.json as trainer answer keys outside input; do not feed them to the model as invoice evidence. AI results can vary, so compare with the source.

Handover checklist

Confirm that the operator can find the input folder; open both local addresses; run a lot; reconcile errors; review the source and entry; choose the right decision; retrieve the correct report; explain email status; record an actual voucher; handle a replacement; and create a backup.

Record during rehearsal: n8n version, Ollama version/model tag, Windows folder, company, reviewer access location and backup location. Run the Outlook sender check separately; do not describe its latest fix as confirmed until it passes on the PC.

QUICK TROUBLESHOOTING

If something does not work

Symptom	Simple next check
5678 will not open	Start Docker Desktop; run docker ps. If the existing n8n container is stopped, use docker start n8n. Confirm no other application owns the port.
8787 will not open	Run START_V2.cmd, then STATUS_V2.cmd. Keep the approval app distinct from n8n; 8787 is its correct port.
Save V2 Extraction: connection closed	Check V2 is running and inspect its status/logs. The message alone does not establish the cause; a successful retry only shows it was recoverable. Check the batch before retrying.
No files received	Check the exact V2 input folder and docker exec n8n ls /files. Use top-level PDFs under 25 MB, not a ZIP or an extra nested folder.
Extraction error or empty text	Try selecting PDF text. Scans need OCR outside this workflow. A PDF containing several invoices must be split; a model change alone does not add OCR.
Qwen/model connection fails	Start Ollama; check ollama list and exact model tag. In n8n use host.docker.internal, not localhost. See the connection note below if Windows works but Docker cannot reach it.
Approval button cannot proceed	Save pending edits, read blocking findings, resolve duplicates and check the accounting balance. Review exceptions need explicit acknowledgment and a reason.
Approved report omits an invoice	It may be reserved in a prior export, changed, not approved or already recorded. Re-download the correct snapshot; do not assume the record was lost.
Split-Path / sender-selection error	Use current Sender Fix 2, close the old helper, open Classic Outlook and run CHECK_OUTLOOK_V2.cmd. Use the exact listed sender. A failed check needs the error details.
Email remains queued / unknown	Check helper and Outlook, refresh the queue, then inspect Sent Items/Outbox/Drafts before any fresh request. A stale request is not automatically retried.

Ollama connection note (installer): if Ollama works on Windows but n8n cannot reach it, check its address, listener and firewall. If needed, quit Ollama, set the Windows user environment variable OLLAMA_HOST to `0.0.0.0:11434`, restart and test. This broadens the listener; restrict access to trusted local/Docker use. [11]

Windows blocks a PowerShell script: show the exact error to the installer or IT and use the approved signing/execution method. Do not weaken the execution policy globally.

Ask for help with: the exact error, failed node, batch/tracking ID and recent relevant logs. Remove private invoice data and tokens before sharing. Do not delete volumes or clear the database as a troubleshooting shortcut.

QUESTIONS TO EXPECT

Trainer FAQs 01–05

01 What do I need before starting, and is this a paid system?

Use a Windows PC with WSL 2 enabled for the Docker Desktop WSL backend, Docker Desktop, a running n8n container, Ollama and the qwen2.5:7b model. SQLite is included, so no separate database purchase or installation is needed; Classic Outlook is optional for email. This local workflow has no per-invoice cloud-model API charge, but hardware, electricity, Outlook and applicable software licensing still have costs or terms; check Docker and n8n terms for your intended commercial use.

02 What is the difference between a Docker image and a container?

An image is the packaged software used to create an application environment. A container is a running instance of that image. Here, n8n runs in one container and the V2 approval application runs in another; restarting one does not mean reinstalling the whole system.

03 What are a folder mount and a Docker volume?

A folder mount makes a Windows folder visible inside a container: your n8n-files folder appears as /files in n8n. A named Docker volume stores persistent application data managed by Docker. V2 keeps its SQLite database in invoice_v2_data, while source PDFs and reports are in the Windows invoice_demo_v2 folder; the database and documents must be backed up together.

04 Why do I use port 5678 for one screen and 8787 for another?

A port identifies a particular application on the same computer. Open http://localhost:5678 to run the n8n workflow and http://localhost:8787 to review invoices in V2. Ollama normally uses port 11434 to receive model requests; it is not the approval screen.

05 What do n8n, Ollama, Qwen and SQLite each do?

n8n carries out the workflow steps, while Ollama runs the Qwen model on your computer. Qwen converts readable invoice text into structured fields and suggests a category. The separate approval application checks and presents those results, and SQLite remembers batches, versions, decisions and export history.

QUESTIONS TO EXPECT

Trainer FAQs 06–10

06 Why does the workflow use host.docker.internal instead of localhost for Ollama?

Inside the n8n container, localhost means that container itself. host.docker.internal lets Docker Desktop reach Ollama running on Windows. The V2 application is reached from n8n through its Docker network name, invoice-approval-v2; browser addresses and container addresses serve different locations.

07 Can I change Qwen to a smaller or larger local model?

The supplied workflow is configured for qwen2.5:7b; use ollama list to check that it is available. Another model may differ in speed, memory use and extraction quality, so change the exact model name in the V2 settings and test a few known invoices before a demonstration. A successful response is not proof that the extracted figures are correct.

08 Do trainees need to install Python, SQLite or learn SQL?

No separate Windows Python or SQLite installation is needed for the supplied package. The V2 container provides Python and SQLite, and setup creates the database automatically. The AP operator works through the approval screen, Excel reports and simple command files.

09 Is everything offline and private?

Invoice extraction and approval run locally when the workflow uses the configured local Qwen model and local endpoints. Initial software and model downloads need internet, and sending email uses your Outlook account and mail service. Local processing does not replace access controls and backups: protect the PC, archives, reports, login file, .env and configured workflow JSON, and avoid uploading real invoices to public AI services.

10 Can ChatGPT or Gemini generate the workflow JSON for me?

Yes, give it the workflow purpose, n8n version, Docker mount, local model name, extraction schema and V2 API contract, using redacted examples. For this existing V2 package, start from its supplied workflow and import the setup-generated Ready_To_Import_V2.json, which has the local connection details. Generated JSON needs an import check and a small test run; ask for specific changes instead of rebuilding a working workflow without its integration details.

QUESTIONS TO EXPECT

Trainer FAQs 11–15

11 Why did a scanned invoice fail when the training PDFs worked?

The current workflow extracts text already contained in a PDF; it does not perform OCR. A scanned page can look readable to a person yet contain no text for n8n to extract. Such a document needs a separate OCR step before this workflow, and the OCR result must be checked against the image.

12 Can one PDF contain several invoices or several pages?

Use one invoice per PDF. A single invoice may span multiple text pages, but review that all relevant lines and totals were captured; the current workflow is not a general document-splitting service. A PDF containing six separate invoices must first be separated into six invoice files, with OCR where needed.

13 If extraction succeeded, why are there still warnings or an approval block?

Extraction success means usable data was received, not that the invoice is correct or ready to book. The system checks matters such as required fields, arithmetic, duplicate candidates and balanced accounting lines. Blocking findings must be resolved; review findings need an explicit decision and reason, supported by the source document.

14 What do the invoice statuses mean?

Processing means extraction is underway; Processing problem means it did not finish successfully; Awaiting approval means it needs review. On hold and Returned for correction remain open for action, while Rejected closed is excluded from outstanding items but retained in history. Approved awaiting entry means the current version is approved but not yet recorded; Manually recorded means the operator has entered a Tally voucher reference. A closed record can be reopened with a reason, but a recorded entry requires separate accounting correction follow-up.

15 What happens if I copy the same invoice into input again?

An identical file is recognised by its content fingerprint and normally skipped, while the receipt is logged; a failed extraction can be retried. A changed file is also checked for a possible duplicate using invoice and supplier details across batches. Review the matching records: duplicate candidates are not automatically accepted as a second liability, and filename changes alone do not make an invoice new.

QUESTIONS TO EXPECT

Trainer FAQs 16–20

16 How do I correct a returned invoice without losing its history?

For an extraction mistake, edit the verified fields or accounting lines with a reason; this creates a new version and requires approval again. For a corrected vendor document, process the new PDF and explicitly link it as the replacement for the outstanding original. Earlier documents and decisions remain available, and any old approved export containing a changed version is marked superseded.

17 Does the system decide the correct Tally ledger, GST credit or TDS treatment?

No: the document supplies invoice facts, while suggested accounting lines are a starting point for the reviewer. Verify the company, actual Tally ledger names, expense-versus-asset treatment and tax treatment before approval; the application does not read your Tally masters or determine tax-credit eligibility. This release supports ordinary INR supplier invoices, not automated TDS, split payables, credit notes, foreign-currency vouchers or inventory posting; keep unsupported cases on hold for separate handling.

18 Why does an approved invoice disappear from the next Approved Entries export?

The first approved export reserves its invoice versions to reduce repeated entry from multiple export files. Re-download that original export when you need another copy. Release its allocation with a reason only when the entries have not been recorded and a fresh export is needed; every workbook is a dated snapshot, so check the current dashboard before booking.

19 Does approval, export or sending an email automatically post to Tally or pay a vendor?

No. The Approved Entries workbook provides a reference for manual Tally entry, and downloading or emailing it does not create a voucher or make a payment. After actual entry, record the company, voucher type, number and date in V2; this is your confirmation, not an independent check against Tally.

20 Can I track pending invoices across lots, users and companies?

Outstanding invoices carry forward across batches, and the database retains earlier versions, reviewer decisions and batch history. A batch report preserves its opening outstanding position and shows current status when the report is generated. This build has one company and one named reviewer account; responsibility can be recorded in remarks, but it is not a multiuser assignment or multi-company system.

QUESTIONS TO EXPECT

Trainer FAQs 21–25

21 Does it download invoices from my mailbox or automatically send approval and rejection messages?

No: the AP operator downloads invoices from email and copies them into the V2 input folder. The operator reviews the results, records reasons, generates the appropriate report and explicitly chooses its Email action and recipients. Approval changes do not automatically send vendor-specific rejection notices, reminders or follow-up calls; the outstanding report and remarks guide that manual follow-up.

22 What must be running before I send a report through Outlook?

Open Classic Outlook and run outlook_bridge\START_OUTLOOK_V2.cmd, leaving the helper window open. Use the exact sender address shown by the helper, select the intended exported report, check To/Cc and submit one request. The email attaches that selected Excel report and explanatory text; it does not automatically attach all reports or all source invoices. Send an Approved Entries report before marking its invoices manually recorded.

23 What do Queued, Submitted, Error and Unknown mean for email?

Queued means the request is waiting for a helper result; Submitted means Outlook accepted the send request, not that the recipient received it. Error means the helper reported a failure; Unknown means the application cannot confirm the result. Requests expire after three minutes and are not automatically retried, so inspect Outlook Sent Items, Outbox and Drafts before creating another request, especially after an unknown outcome.

24 What should I do about the Split-Path or requested sender account error?

Install the latest supplied Sender Fix 2 files into the existing outlook_bridge folder; this includes the earlier startup-path correction. Run CHECK_OUTLOOK_V2.cmd with Classic Outlook open and look for PASS for your intended sender; this check does not send email or process the queue. The updated sender fix still needs confirmation on your PC, and a PASS verifies account selection only, not mail-server connectivity or recipient delivery.

25 Will restarting lose my data, and how do I recover after a failure?

Normal container restart preserves records in the Docker volume and files in the mounted folder; deleting those stores can lose data. Use BACKUP_V2.cmd while V2 is running, keep a copy of the ZIP on another disk and keep your private configuration separately; the backup includes the database, archive and output, not just the invoice list. If processing was interrupted, inspect the batch and errors, then close the interrupted batch with a reason before retrying after the fault is fixed. Restore only through the supplied restore procedure, with the helper and V2 workflow stopped, and check Outlook before any fresh email request afterward.

REFERENCE

Terms and trusted references

Term	Plain meaning
Batch / lot	One workflow run and its received-file log.
Tracking ID / version	The stable invoice case number / one saved state of its data.
Exception / block	A finding to review / a condition that prevents approval.
Snapshot / reservation	A report captured at a point in time / versions allocated to an approved export.
Mount / named volume	A Windows folder visible to Docker / persistent storage managed by Docker.
Local model / OCR	AI running on this PC / converting an image into machine-readable text.
JSON / API	A structured text format used for the workflow file / the interface applications use to exchange data.

Official installation and product references

[1] Microsoft: install WSL

[2] Docker Desktop: Windows installation

[3] Ollama: Windows setup

[4] Ollama: qwen2.5:7b model

[5] n8n: import and export workflows

[6] SQLite: copyright and public-domain status

[7] Docker Desktop: licence terms

[8] n8n: Sustainable Use License

[9] n8n Docker Hub image

[10] Docker Desktop: networking

[11] Ollama: FAQ and host configuration

[12] Docker Desktop: connect to host services

Basis and release boundary

Application instructions were checked against the supplied V2 workflow, Python application, browser interface, setup/backup scripts, sample-pack answer keys and Outlook Sender Fix 2. Official setup references were checked on 9 October 2026. Menus in later software versions may move.

The guide describes the current local training/control workflow. It does not claim unattended vendor communications, an enforced multiuser approval matrix, tamper-proof audit storage, OCR or automatic Tally posting. The latest Outlook patch remains subject to verification on the actual Windows desktop.